Analyzing and Improving FlowMap Distillation of Autoregressive Video Models

Anonymous CVPR submission

Paper ID *****

Abstract

Few-step autoregressive (AR) video diffusion models currently rely on self-forcing distribution matching distillation (SF-DMD), which mitigates rollout drift but reduces sample diversity and requires a large bidirectional teacher. FlowMap distillation is a natural alternative that preserves diversity and distills an AR teacher directly, yet existing AR FlowMap students still depend on a final SF-DMD stage to be competitive. In this paper, we identify and address two obstacles that limit causal FlowMap distillation across its teacher-forcing and self-forcing stages. First, in teacher forcing, the student struggles to learn motion because high uncertainty in moving regions causes the teacher's velocity field to change sharply. Second, in self-forcing, the student is supervised on its own rollouts by an AR teacher, which simply continues a drifted history rather than recovering from it. We address each obstacle by introducing motion-aware guidance*, which applies stronger skip-layer guidance to high-uncertainty moving regions, and* anti-drifting guidance*, which pushes the teacher's velocity away from the drift direction estimated by a drift model trained on the student's rollouts. Because both only modify the teacher's velocity field, they apply directly to any FlowMap objective. Applied to both π-Flow and consistency distillation with 1.3B AR teachers, our guidance consistently improves generation quality and reduces drift over 30 s rollouts; in particular, our 4-step π-Flow students match SF-DMD students distilled from 14B teachers on VBench while maintaining substantially higher sample diversity.*

1. Introduction

Video generation models now produce realistic videos [4, 14, 16, 23, 41]. In particular, autoregressive (AR) video diffusion models generate a video chunk by chunk, conditioning each chunk on the ones before it [17, 35, 49]. A video can thus be streamed while it is generated, and the model can respond to user input along the way, which enables real-time interaction and world simulation [25, 37, 45]. Real-time interaction, however, requires each chunk to be generated quickly, whereas a diffusion model takes many denoising steps per chunk, so AR video models are distilled into students that generate each chunk in a few steps.

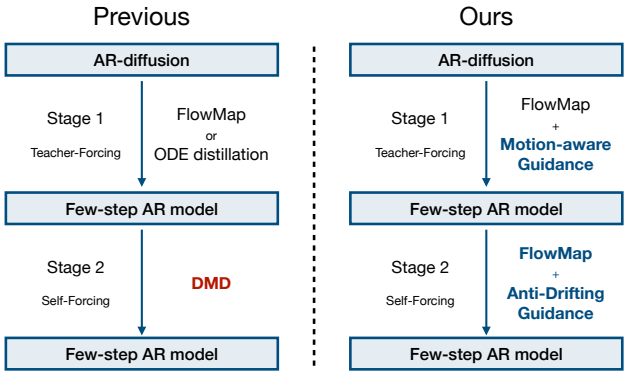


Figure 1. **Two-stage AR video distillation.** Previous methods initialize a few-step student with teacher forcing and then use DMD under self-forcing. We use FlowMap distillation in both stages, with motion-aware guidance in teacher forcing and additional anti-drifting guidance in self-forcing.

The de facto standard recipe for this distillation was introduced by Huang et al. [17] and Zhu et al. [57] (Figure 1, left). Training proceeds in two stages: in Stage 1 (teacher forcing), the AR student learns few-step jumps on clean histories from an AR teacher via FlowMap or ODE distillation; in Stage 2 (self-forcing), the student is trained on its own rollouts with distribution matching distillation (DMD) [47, 48] using a bidirectional teacher. This second stage, which we refer to as self-forcing DMD (SF-DMD), is effective at mitigating rollout drift [17, 40, 58] and has been adopted by most recent AR video models [27, 29, 45, 54]. DMD, however, comes at a cost: it reduces the diversity of the generated videos [8, 11, 36], and the score it matches comes from a large bidirectional model that does not share the causal structure of AR generation [10].

The natural alternative is FlowMap distillation, which trains the student to jump directly along the teacher's denoising trajectories [5, 7, 12, 13, 22, 24, 39]. FlowMap distillation preserves the teacher's distribution, can distill an



(a) Motion-aware guidance (b) Anti-drifting guidance

Figure 2. **FlowMap distillation with and without our guidance.** (a) 5 s videos from π-Flow students after teacher forcing with CFG and with motion-aware guidance. (b) 30 s videos from π-Flow students without and with anti-drifting guidance. Rows in each panel share the prompt and noise, and all students generate with 4 NFE.

AR teacher directly, and rivals DMD in quality on bidirectional video models while generating more diverse videos [36]. Early attempts have applied it to AR video [53, 54], but their students fall far behind SF-DMD students and still rely on a final SF-DMD stage to become competitive. So far, DMD has remained necessary to distill AR video models. In this paper, we show that it is not.

We hypothesize that the gap between FlowMap distillation and DMD for AR video stems from at least two factors. First, motion is hard for a FlowMap student to learn. A FlowMap student has to follow the teacher's velocity field with a few long jumps, whereas DMD only matches the distribution of its outputs and is not burdened with following trajectories [47]. Where the clean endpoint is uncertain, the teacher's velocity field changes sharply across nearby states at the same noise level, making a few-step student sensitive to small approximation errors [30, 51]. In AR video, this uncertainty is concentrated: after the first chunk, the history already determines how the scene looks, so the remaining uncertainty lies in how it moves. The errors of a FlowMap student therefore concentrate in moving regions (Figure 2(a)). We demonstrate that while guidance can smooth the teacher's velocity field, this effect is pronounced in the first chunk yet marginal in later chunks (Section 4), as classifier-free guidance [15] only strengthens the text condition, which the history already largely satisfies. We therefore propose *motion-aware guidance*, which applies skip-layer guidance [19, 36] at each patch with a strength set from the uncertainty that the teacher predicts there.

Second, an AR teacher cannot recover from a drifted history in Stage 2. To mitigate exposure bias, self-forcing trains the student on its own rollouts, which gradually drift away from real videos. SF-DMD matches the joint distribution of the full video sequence $p(x^{1:L})$ using a bidirectional teacher, which directly penalizes sequence-level drift. Causal FlowMap distillation, by contrast, matches the AR teacher's conditional distribution $p(x^i \mid \hat{x}^{<i})$ given the student's rollout history $\hat{x}^{<i}$. Because the AR teacher treats $\hat{x}^{<i}$ as a fixed condition and was trained only on clean histories, its targets simply continue the drifted history while the few-step student drifts further outward on each chunk (Figure 2(b)). We therefore propose *anti-drifting guidance* (Section 5), which trains an additional causal model on the student's rollouts to capture these further-drifted transitions on the same history and uses it as a negative branch to push the teacher's velocity back toward the data distribution [20].

Because both guidance methods modify only the teacher's velocity field, they apply directly to any FlowMap objective (Figure 1, right). We instantiate them with two representative FlowMap methods—π-Flow [7] and discrete-time consistency distillation (dCM) [39, 54]—across both chunk-wise and frame-wise AR teachers from Causal Forcing [57] and Causal-rCM [54] on Wan2.1-1.3B [41]. Distilled from 1.3B AR teachers without any DMD stage, our 4-step FlowMap students match SF-DMD students distilled from 14B teachers on VBench while generating substantially more diverse videos and mitigating drift in 30-second rollouts.

2. Related Work

Few-step AR video diffusion AR video diffusion models generate chunks sequentially with causal attention and a KV cache [6, 35, 49]. To mitigate exposure bias from chunk-to-chunk error accumulation at inference, most few-step AR models train the student on its own rollouts with SF-DMD [17, 53, 54, 57] or an adversarial loss [27], while subsequent works modify DMD to recover the diversity it loses [8, 11, 52]. Beyond rollout training and long-horizon gradient routing [9, 40, 58], extrapolating past the training context window also causes KV-cache drift, which recent methods address through KV recaching, attention sinks, and KV-cache compression [29, 45, 46]. Cache management operates at inference and complements our approach, whereas our work replaces the DMD distillation objective itself with FlowMap distillation.

FlowMap distillation Consistency models map any noisy state to the clean endpoint of its trajectory [39] and are trained in discrete time [32, 42] or in continuous time with Jacobian–vector products (JVPs) [31, 55]. FlowMap distil-

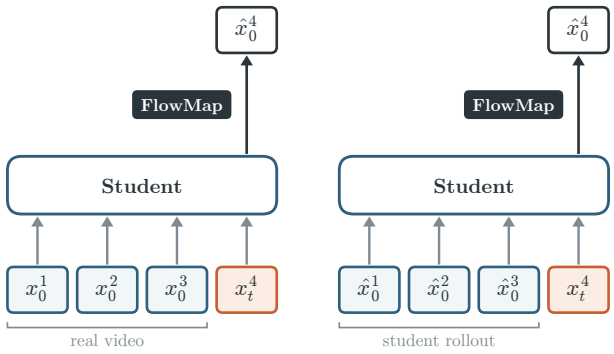
lation generalizes this endpoint mapping to jumps between any two noise levels [5, 12, 13, 22, 34, 56]. π-Flow [7] parameterizes the flow map through a velocity policy without JVPs, and PDD [36] scales this approach to large bidirectional video models with a discrete-velocity policy. Recent works have applied consistency distillation to causal AR video models [53, 54], but only as a teacher-forced warm-up or regularizer before falling back to SF-DMD. We identify and address the failure modes of standalone causal FlowMap distillation across both teacher forcing and self-forcing.

Guidance Beyond classifier-free guidance (CFG) [15], which contrasts conditional and unconditional predictions, guidance can construct a weaker negative branch by perturbing self-attention [1], skipping layers [19], or querying a weaker version of the model [20]. Whereas prior skip-layer guidance applies a uniform strength across the frame, our motion-aware guidance scales patch-wise skip strength from the teacher's predicted uncertainty to smooth the velocity field in moving regions. To reduce drift during multi-step AR sampling, recent works also propose anti-drifting frame packing [50] and history guidance [26, 38], which require specialized training or extra inference evaluations. Most closely related to our anti-drifting guidance, Alemohammad et al. [2] fine-tune a copy of a diffusion model on its own synthetic data and use it as a negative guide during static sampling; in contrast, we update an online drift model on the student's rollouts during self-forcing distillation to steer causal AR transitions back toward the data distribution.

3. Preliminaries

AR video diffusion We write a video latent as a sequence of chunks $x = (x^1, \dots, x^L)$, each consisting of a few latent frames (three in our chunk-wise setting and one in our frame-wise setting). An AR diffusion model factorizes $p(x) = \prod_i p(x^i \mid x^{<i})$ and denoises each chunk with causal attention to its history cached in a KV cache. With noise level $t \in [0, 1]$, flow matching [28, 30] perturbs a clean chunk $x_0^i \equiv x^i$ as $x_t^i = (1-t)x_0^i + t\epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$ and trains a network on clean histories to predict the velocity $\epsilon - x_0^i$ [57]. We call this model the teacher and write $v_\phi(x_t^i, t \mid h_i)$ for its conditional velocity given history h_i (omitting the text prompt for brevity). To sample, the teacher denoises each chunk from $t = 1$ to $t = 0$ with a few tens of ODE steps using a guided velocity $v^\star$ and appends the generated chunk to the history.

Two-stage AR distillation A few-step AR student generates each chunk in a few steps and appends it to the history of the next chunk. Distillation proceeds in two stages



(a) Teacher forcing (b) Self-forcing

Figure 3. **Teacher forcing and self-forcing.** The student generates the chunk $\hat{x}_0^4$ from the noisy chunk x_t^4 given the history. (a) In teacher forcing, the history is real video. (b) In self-forcing, the history is the student's own rollout.

(Figure 3): Stage 1, *teacher forcing* (TF), conditions every chunk on the clean history $h_i = x_0^{<i}$ from training data, whereas Stage 2, *self-forcing* (SF) [17], conditions chunk i on the student's own rollout history $h_i = \hat{x}_0^{<i}$ to match inference. While standard recipes switch to DMD in Stage 2 (SF-DMD), we use FlowMap distillation in both stages.

FlowMap distillation Given a history h_i, the teacher's flow map transports chunk i from noise level s to a lower level t along the probability-flow ODE,

$$\Phi_{t\leftarrow s}(x_s^i \mid h_i) = x_s^i - \int_t^s v^\star(x_\tau^i, \tau \mid h_i)\, d\tau.$$

FlowMap distillation trains a student map $\Phi_{t\leftarrow s}^\theta$ to approximate this transition in a few steps. Following Boffi et al. [5], FlowMap objectives fall into two main families—Lagrangian self-distillation (LSD) and Eulerian self-distillation (ESD)—depending on whether they differentiate the flow map with respect to the target time t or the source time s. As an instance of LSD, we use *π-Flow* [7, 36], which matches the target-time derivative of the student map to the teacher velocity at the state reached by the student's jump (parameterization details in the supplementary material):

$$\begin{aligned} \mathcal{L}_{\text{LSD}} &= \mathbb{E}\big\|\partial_t \Phi_{t\leftarrow s}^\theta(x_s^i \mid h_i) - v^\star(\hat{x}_t^i, t \mid h_i)\big\|^2, \\ \hat{x}_t^i &= \text{sg}\big[\Phi_{t\leftarrow s}^\theta(x_s^i \mid h_i)\big], \end{aligned} \tag{1}$$

where $\text{sg}[\cdot]$ denotes stop-gradient. As a finite-difference instance of ESD, we use *discrete-time consistency distillation* (dCM) [39, 54], which trains an endpoint map $f_\theta(x_s^i, s \mid h_i) = \Phi_{0\leftarrow s}^\theta(x_s^i \mid h_i)$ by enforcing invariance across a

teacher ODE step from s to $t < s$:

$$\mathcal{L}_{\text{ESD}} = \mathbb{E}\big\| f_\theta(x_s^i, s \mid h_i) - \text{sg}\big[f_\theta(\tilde{x}_t^i, t \mid h_i)\big] \big\|^2, \quad \tilde{x}_t^i = x_s^i - (s-t)\, v^\star(x_s^i, s \mid h_i). \tag{2}$$

Guidance Guidance extrapolates the teacher's conditional prediction away from a weaker negative branch v_ϕ^- as $v^\star = v_\phi + (w-1)(v_\phi - v_\phi^-)$ with scale $w > 1$, where v_ϕ^- can be obtained using a negative prompt (CFG [15]), skipping network blocks (skip-layer guidance, SLG [19]), or querying a weaker model (autoguidance [20]). Crucially, in both LSD (Eq. (1)) and ESD (Eq. (2)), the teacher enters the objective exclusively through its guided velocity $v^\star$, and distillation absorbs this guidance into the student so that no guidance is needed at inference.

4. Stage 1: Teacher Forcing

4.1. Where Teacher Forcing Fails

Because neural networks are inductively biased toward smooth, low-Lipschitz functions [33], a distillation target is easier for a few-step student to learn when it varies smoothly across states. In FlowMap distillation, however, the student must learn the time-integrated flow map $\Phi_{t \leftarrow s}(x_s^i \mid h_i)$, whose spatial Lipschitz constant grows exponentially with the local Lipschitz constant of the teacher's velocity field $v^\star(x_\tau^i, \tau \mid h_i)$ across noise levels $\tau \in [t, s]$, bounded by $\exp\big(\int_t^s \|\nabla_x v^\star(\cdot, \tau)\|\, d\tau\big)$ (Proposition 1 in the supplementary material). Keeping the local Lipschitz constant of $v^\star$ small at each noise level t is therefore essential to make FlowMap distillation easy to learn.

A natural way to smooth the teacher's velocity field is guidance. We quantify local sensitivity across nearby states x_a and x_b at the same noise level by

$$C_v = \frac{\|v^\star(x_a) - v^\star(x_b)\| \,/\, \|v^\star\|}{\|x_a - x_b\| \,/\, \|x\|}, \tag{3}$$

and plot the relative gain in smoothness $1/C_v$ over the unguided teacher ($\lambda = 0$) across normalized guidance magnitudes λ in Figure 4 (protocol in the supplementary material). In the first chunk, CFG effectively smooths the velocity field, raising $1/C_v$ by up to 7.7% (Figure 4(a)). In the second chunk, however, CFG fails—raising $1/C_v$ by at most 0.3% and lowering it as λ grows (Figure 4(b)).

Why does CFG stop working after the first chunk? Under flow matching, the local Lipschitz sensitivity of the teacher's velocity is governed by posterior uncertainty—equivalently, the expected squared velocity residual $\mathbb{E}[\|r\|^2 \mid x_t^i, h_i]$ with $r = v_\phi(x_t^i, t \mid h_i) - (\epsilon - x_0^i)$ (Proposition 1). In the first chunk, the entire scene is uncertain and generated from scratch under the text prompt. Once the first chunk is generated, however, the history

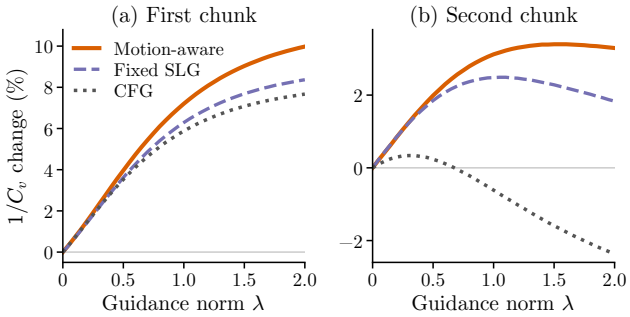


Figure 4. **Velocity smoothness $1/C_v$ across chunks and guidance methods.** Percent change in $1/C_v$ (Eq. (3)) over the unguided teacher ($\lambda = 0$) as a function of the normalized guidance magnitude λ, pooled over 20 noise levels and 64 clips for the Causal Forcing chunk-wise teacher. (a) In the first chunk, all three guidance methods smooth the velocity field. (b) In the second chunk, CFG fails to increase $1/C_v$ and degrades it at larger λ, fixed-strength skip-layer guidance (Fixed SLG) saturates early, and motion-aware guidance achieves the largest gain.

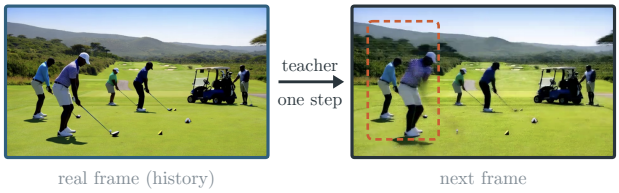


Figure 5. **One-step teacher sample.** Given a real frame as history (left), the teacher generates the next frame in one step from noise (right). The dashed box marks the moving player.

h_i already fixes the static background and largely satisfies the text prompt, so uncertainty—and thus high Lipschitz sensitivity—concentrates in *moving regions*: as shown by the 1-NFE teacher prediction in Figure 5, a single step from noise already recovers a sharp background while leaving the moving player blurred. Because CFG only strengthens the text condition, it cannot guide this frame-to-frame motion; meanwhile, although skip-layer guidance (SLG) [19] does not rely on the text prompt, applying a uniform skip strength across all patches (Fixed SLG) perturbs static and moving regions alike, saturating early (+2.5%; Figure 4(b)).

4.2. Motion-Aware Guidance

To guide motion-related regions, we introduce *motion-aware guidance*, which sets the patch-wise skip-layer strength from the teacher's predicted uncertainty (Figure 6). First, we train a lightweight head on the detached features of block ℓ of the frozen teacher with the Gaussian negative log-likelihood [21] to predict the uncertainty $\hat{u}_p$ of each patch p from the flow-matching residual r. At guidance time, we spatially smooth $\hat{u}_p$ with a 3×3 mean pool and form

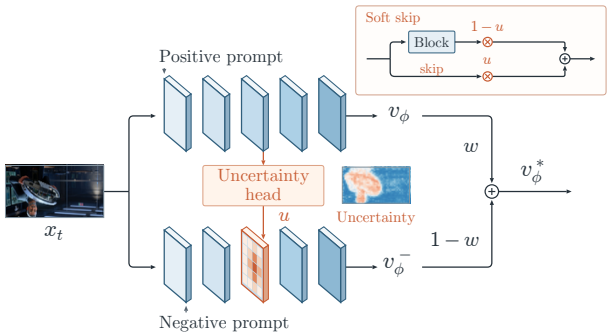


Figure 6. **Motion-aware guidance.** Uncertainty predicted from the conditional teacher's features determines spatially varying skip strengths u through smoothing and normalization (Eq. (4)). The map illustrates the predicted uncertainty, and the inset shows the soft-skip operation. The resulting weak prediction v_ϕ^- is combined with v_ϕ to obtain the guided velocity $v^\star$ (v_ϕ^* in the diagram). The branches share the noisy chunk, noise level, and history; the frame illustrates the current chunk.

$v^\star = v_\phi + (w-1)\,(v_\phi - v_\phi^-)$, where the negative branch v_ϕ^- uses the negative prompt and skips block ℓ with patch-wise strength u_p:

$$z = z^{\text{no-skip}} + u \odot (z^{\text{skip}} - z^{\text{no-skip}}),$$
$$u_p = \max\Big(0,\ \bar{u} + \sigma_u \frac{\hat{u}_p - \mu}{\sigma}\Big). \tag{4}$$

Here $z^{\text{no-skip}}$ and z^{skip} denote the output and input of block ℓ ($u=0$ recovers CFG and $u=1$ skips the block), (μ, σ) are the mean and standard deviation of $\hat{u}_p$ tabulated for each chunk index and noise level t to normalize the prediction, and $\bar{u}$ and σ_u are hyperparameters that set the base strength and its spatial variation. With this design, motion-aware guidance raises whole-chunk smoothness $1/C_v$ in the second chunk monotonically by $+3.4\%$ (Figure 4(b)) and local smoothness on the top uncertainty decile by $+16.4\%$ over CFG (supplementary material).

5. Stage 2: Self-Forcing

5.1. Why Self-Forcing Fails

A student trained only on clean histories suffers from exposure bias when rolled out autoregressively, so Stage 2 (self-forcing) trains the student on its own generated histories $h_i = \hat{x}_0^{<i}$ (Figure 7a). In SF-DMD, a bidirectional teacher noises and scores the entire rollout to match the joint sequence distribution $p(x^{1:L})$; because this teacher does not condition on the history and instead supervises the full sequence jointly, it directly penalizes sequence-level drift. In causal FlowMap distillation, however, the student matches the conditional distribution $p(x^i \mid \hat{x}_0^{<i})$ of the AR teacher $v_\phi(x_t^i, t \mid \hat{x}_0^{<i})$, which takes the student's rollout as a fixed condition through its KV cache. Because the teacher was trained only on clean histories, nothing in its training asks it to repair a corrupted history: given a drifted history, it continues parallel to the ground-truth distribution while retaining the history's offset (Figure 7c, Teacher). Matching the teacher's continuation therefore preserves existing errors rather than pulling the rollout back toward the data distribution.

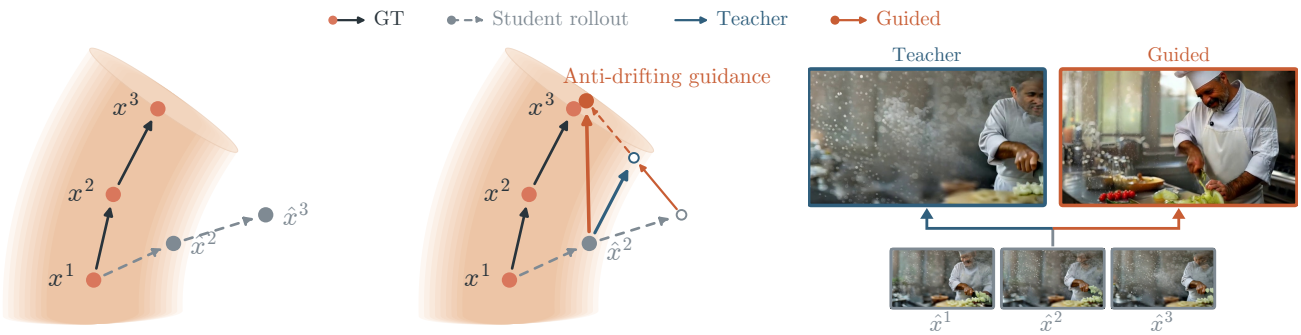
5.2. Anti-Drifting Guidance

To obtain a restoring direction, we guide the teacher in the direction opposite to the student's drift. We capture the student's rollout transitions with a drift model v_ψ, a trainable copy of the teacher fit with flow matching to the student's generated chunks $\hat{x}_0^i$ on the same rollout history $\hat{x}_0^{<i}$. Using the original teacher v_ϕ as the positive branch and the drift model v_ψ as the negative branch, we add *anti-drifting guidance* to form the full guided velocity:

$$v^\star = v_\phi + (w-1)\,(v_\phi - v_\phi^-) + (\gamma - 1)\,(v_\phi - v_\psi), \tag{5}$$

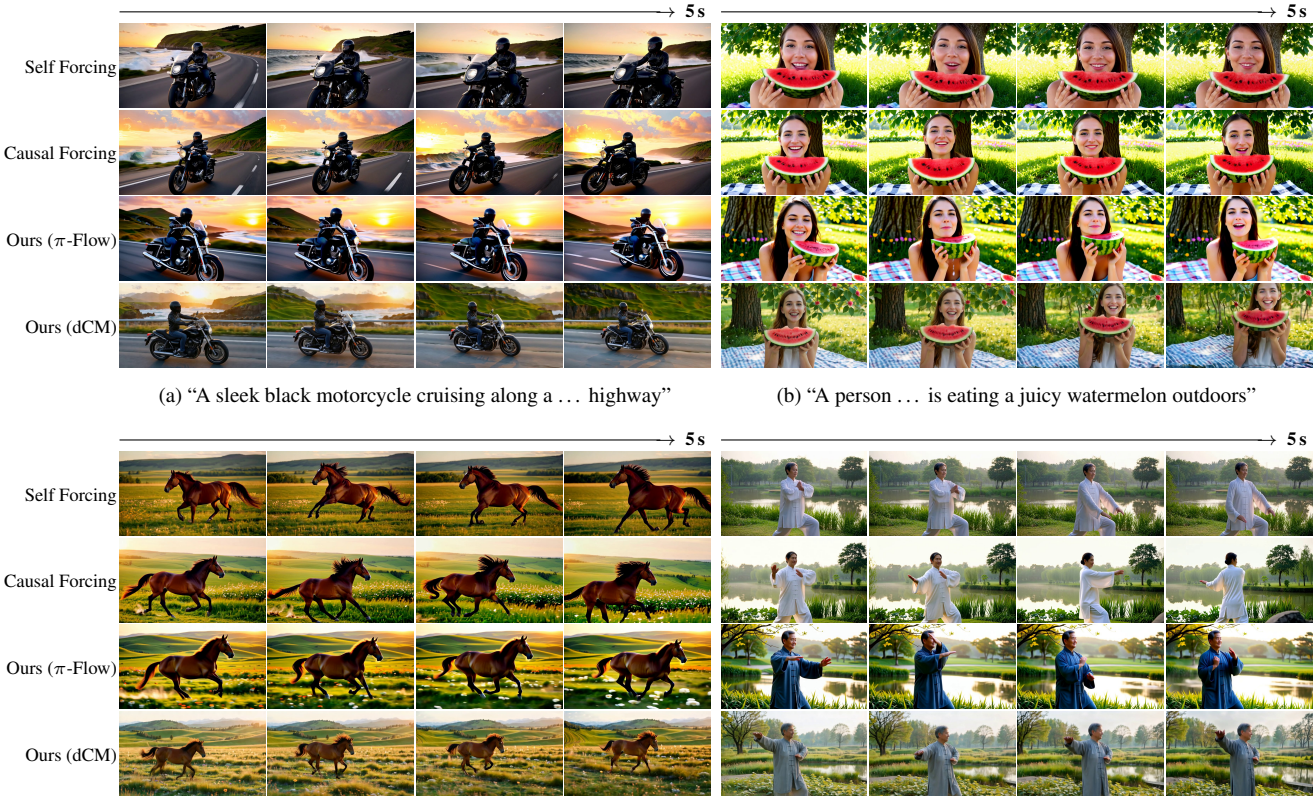
where v_ψ uses the positive prompt at the same state and history as v_ϕ, and anti-drifting guidance is applied only in Stage 2 after the first chunk ($\gamma = 1$ in Stage 1 and on the first chunk). Because v_ϕ continues parallel to the data distribution while v_ψ points along the further-drifted student rollout, their difference $v_\phi - v_\psi$ points inward, opposite to the per-chunk drift, and scaling by $\gamma > 1$ extrapolates this inward correction to steer the guided transition back toward the data distribution (Figure 7b, green arrow; Figure 7c, Guided). Because the student changes at every update in Stage 2, we update v_ψ online for n steps per student update on the student's rollouts.

Difference from DMD Although anti-drifting guidance and DMD appear similar in contrasting the teacher with a model trained on student rollouts, they are fundamentally different. DMD minimizes the mode-seeking reverse KL divergence $D_{\text{KL}}\big(q(x_0^{1:L}) \,\|\, p(x_0^{1:L})\big)$ at the clean output by re-noising the student's endpoint $\hat{x}_0^{1:L}$ with fresh noise and backpropagating the score difference directly into $\hat{x}_0^{1:L}$, which discards the noise-to-data trajectory coupling. By contrast, anti-drifting guidance preserves the deterministic coupling of FlowMap distillation and modifies only the causal teacher's velocity field at the intermediate noisy state $x_t^i \mid \hat{x}_0^{<i}$ along the denoising trajectory: adding $(\gamma-1)(v_\phi - v_\psi)$ shifts the denoising direction at x_t^i in proportion to the log density ratio $\nabla_{x_t^i} \log \frac{p(x_t^i \mid \hat{x}_0^{<i})}{q(x_t^i \mid \hat{x}_0^{<i})}$, steering the guided trajectory toward $p^\star(x_t^i \mid \hat{x}_0^{<i}) \propto p(x_t^i \mid \hat{x}_0^{<i}) \big(\frac{p(x_t^i \mid \hat{x}_0^{<i})}{q(x_t^i \mid \hat{x}_0^{<i})}\big)^{\gamma-1}$. Indeed, empirically, our anti-drifting guidance increases sample diversity rather than collapsing modes (Table 6).



(a) Exposure bias (b) Anti-drifting guidance (c) Teacher given a student rollout

Figure 7. **Anti-drifting guidance.** Shading represents high-probability regions of the data distribution. (a) Student rollouts drift from the ground-truth (GT) trajectory. (b) From $\hat{x}^2$, the teacher continues parallel to GT but retains the offset. Adding the difference between the teacher and rollout directions brings the guided transition closer to GT. The dashed orange arrow is a translated copy of this correction. Arrows illustrate chunk transitions. (c) Given a student rollout as history, the teacher generates the next chunk without (Teacher) and with (Guided) anti-drifting guidance.



(a) "A sleek black motorcycle cruising along a … highway" (b) "A person … is eating a juicy watermelon outdoors"

(c) "A majestic brown horse galloping across … field" (d) "A person practicing Tai Chi in a serene park …"

Figure 8. **Qualitative comparison of 4-NFE chunk-wise students.** Frames at 0, 1.7, 3.4 and 5 s of videos from Self Forcing, Causal Forcing, ours (π-Flow) and ours (dCM) with the Causal Forcing teacher, for four VBench motion prompts (abbreviated) and the same seed.

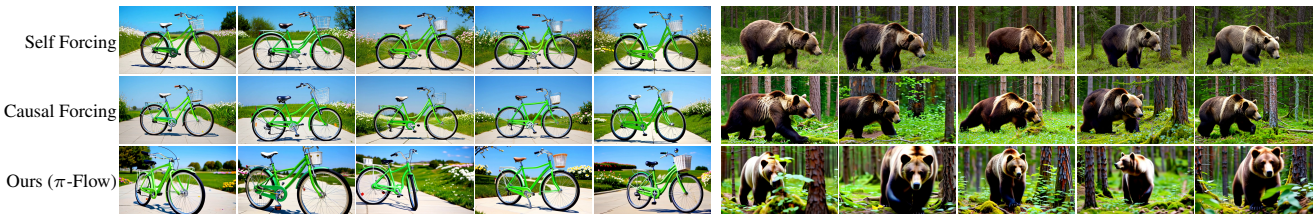
6. Experiments

6.1. Setup

Teachers and guidance We distill the AR teachers released by Causal Forcing [57] and Causal-rCM [54]. Both

Table 1. **Comparison of 4-NFE AR students on Wan2.1-1.3B.** VBench scores and dynamic degree are five-seed means; the V-JEPA 2 and VideoMAE V2 columns measure diversity as the mean pairwise feature distance among five videos per prompt (higher is more diverse). Init cites the work that released the causal 1.3B checkpoint that initializes the student (for the AR teachers, the evaluated checkpoint), Type is the distillation of its final stage, and Teacher is the model it is distilled from. All models are Wan2.1 [41]; a superscript AR marks a causal model trained with diffusion under teacher forcing, and the others are bidirectional. †Our evaluation of the official checkpoint. Results are ranked separately within the chunk-wise and frame-wise settings and colored from dark to light: best, second, third and the rest; the AR teachers are shown in gray for reference and are not ranked.

	Init	Method	Type	Teacher	NFE	VBench ↑ Total	Quality	Semantic	Dynamic	V-JEPA 2 ↑ Cos	L2	VideoMAE V2 ↑ Cos	L2
Chunk-wise	[17]	Self Forcing†	DMD	14B	4	84.22	84.83	81.81	66.1	0.0555	21.5	0.0117	2.50
	[57]	AR teacher, CFG	–	–	32 × 2	84.46	85.11	81.86	77.8	0.0882	26.9	0.0236	3.50
		Causal Forcing†	DMD	14B	4	84.74	85.41	82.02	83.9	0.0745	24.4	0.0198	3.24
		Ours (π-Flow)	FlowMap	$1.3B^{AR}$	4	84.76	85.83	80.46	88.3	0.0964	27.8	0.0281	3.81
		Ours (dCM)	FlowMap	$1.3B^{AR}$	4	83.70	84.55	80.32	69.7	0.0875	26.6	0.0241	3.54
	[54]	AR teacher, CFG	–	–	32 × 2	84.38	84.85	82.47	65.0	0.0624	23.2	0.0128	2.60
		Causal-rCM†	DMD	$14B^{AR}$, 14B	4	84.92	85.44	82.82	81.1	0.0666	23.6	0.0166	2.95
		Causal-rCM, TF-sCM†	FlowMap	$14B^{AR}$	4	83.78	84.49	80.92	67.5	0.0789	26.2	0.0182	3.13
		Ours (π-Flow)	FlowMap	$1.3B^{AR}$	4	84.87	85.76	81.29	71.4	0.0735	25.2	0.0143	2.77
		Ours (dCM)	FlowMap	$1.3B^{AR}$	4	84.54	85.20	81.91	68.3	0.0634	23.4	0.0122	2.56
Frame-wise	[57]	AR teacher, CFG	–	–	32 × 2	82.84	83.38	80.68	88.6	0.0944	27.4	0.0306	4.01
		Causal Forcing†	DMD	14B	4	82.79	83.52	79.87	98.6	0.0514	19.9	0.0153	2.81
		Causal Forcing++† [53]	DMD	14B	2	84.55	85.25	81.75	63.9	0.0608	22.5	0.0102	2.33
		Ours (π-Flow)	FlowMap	$1.3B^{AR}$	4	83.35	84.48	78.83	64.7	0.0800	26.1	0.0179	3.11
		Ours (dCM)	FlowMap	$1.3B^{AR}$	4	82.97	83.96	78.99	68.9	0.0940	27.4	0.0276	3.77
	[54]	AR teacher, CFG	–	–	32 × 2	83.84	84.51	81.19	77.8	0.0659	23.6	0.0165	2.95
		Causal-rCM†	DMD	$14B^{AR}$, 14B	4	84.46	85.26	81.27	67.5	0.0670	23.9	0.0117	2.53
		Causal-rCM, TF-sCM†	FlowMap	$14B^{AR}$	4	82.94	84.12	78.21	71.4	0.0856	27.3	0.0216	3.45
		Ours (π-Flow)	FlowMap	$1.3B^{AR}$	4	84.04	85.10	79.80	73.9	0.0733	25.0	0.0185	3.11
		Ours (dCM)	FlowMap	$1.3B^{AR}$	4	83.78	84.64	80.33	60.8	0.0715	24.7	0.0148	2.75



(a) "A green bicycle" (b) "A bear hunting for prey"

Figure 9. **Five videos per prompt.** The frame at 2.5 s of seeds 0–4 from Self Forcing, Causal Forcing and ours (π-Flow) with the Causal Forcing teacher. The V-JEPA 2 cosine distance among the five videos is (a) 0.021, 0.078 and 0.127 and (b) 0.032, 0.037 and 0.079, respectively.

are causal Wan2.1-T2V-1.3B models [41] trained with diffusion under teacher forcing on synthetic videos from Wan2.1-T2V-14B. The chunk-wise teachers generate 81-frame videos at 832 × 480 in seven chunks of three latent frames, and the frame-wise teachers generate one latent frame per chunk. We sample the teachers with 32 Euler steps and a timestep shift of 6. Motion-aware guidance (Eq. (4)) uses $w = 5$, block $\ell = 10$, $\bar{u} = \sigma_u = 0.5$, and 3×3 spatial mean pooling of $\hat{u}_p$; its population statistics are tabulated at 64 noise levels and interpolated linearly in between, and the uncertainty head is trained beforehand for 500 updates with a batch size of 8. Anti-drifting guidance (Eq. (5)) uses $\gamma = 5$.

Distillation All students are initialized from their respective AR teachers and generate each chunk in 4 NFE without guidance at inference. Stage 1 trains the student under teacher forcing on clean video latents generated by Wan2.1-T2V-14B [41], and Stage 2 trains it under self-forcing on the extended VidProM prompts [17, 44]. In Stage 2, the drift model is initialized from the teacher and updated $n = 5$ times per student update on the student's rollouts. Method-



(a) "… a grandmother … behind a colorful birthday cake" (b) "… a panda playing the guitar …"

Figure 10. **30 s generation.** Frames at 0, 10, 20 and 30 s of videos from Self Forcing, Causal Forcing, ours (π-Flow) and ours (dCM) with the Causal Forcing teacher, generated with Deep Forcing [46] from the same prompt and seed (prompts abbreviated).

specific formulations and hyperparameters for π-Flow and dCM are provided in the supplementary material.

Baselines We compare with four SF-DMD students: Self Forcing [17], Causal Forcing [57], Causal Forcing++ [53], and Causal-rCM [54]. Causal-rCM releases AR teachers that we use but distills its own students from 14B teachers, with DMD against a 14B bidirectional teacher. It also releases the consistency-distilled student obtained with teacher forcing before SF-DMD (TF-sCM), which we report as a FlowMap baseline. Rows marked † are our evaluations of official checkpoints with their official inference settings.

Evaluation We report the VBench [18] Total, Quality, and Semantic scores and its dynamic degree on the 946 extended prompts of Self Forcing [17], averaged over five seeds. Following PDD [36], we measure diversity across all 946 prompts as the mean pairwise distance between the features of five videos per prompt, using V-JEPA 2 [3] and VideoMAE V2 [43] with cosine ($1 - \cos$) and L2 distances. For long videos, we generate 30 s videos for the 128 MovieGen prompts of Deep Forcing [46] with its KV-cache management (Deep Sink and participative compression), and split each into 16 clips of about 2 s; ΔIQ and ΔAQ are the mean VBench imaging and aesthetic quality of the last three clips minus those of the first three, averaged over five seeds.

6.2. Results

5 s video generation Table 1 compares the 4-NFE students on 5 s generation, and Figure 8 shows qualitative examples. In the chunk-wise setting, our π-Flow students match the SF-DMD students across both teacher families without any DMD stage: ours (π-Flow) reaches 84.76 VBench with the Causal Forcing teacher (vs. 84.74 for Causal Forcing) and 84.87 with the Causal-rCM teacher (vs. 84.92 for Causal-rCM, which is distilled from 14B teachers), while achieving the highest VBench Quality scores overall (85.83 and 85.76) and outperforming both 64-NFE AR teachers; ours (dCM) also surpasses both the 64-NFE teacher and TF-sCM with the Causal-rCM teacher (84.54 vs. 84.38 and 83.78). Compared with the CFG-only π-Flow baseline without our guidance, our guidance raises VBench by +3.15 points with the Causal Forcing teacher (81.61 to 84.76; Table 3) and +0.79 points with the Causal-rCM teacher (84.08 to 84.87; supplementary material). In the frame-wise setting, both ours (π-Flow) and ours (dCM) outperform the 4-NFE Causal Forcing student (83.35 and 82.97 vs. 82.79) as well as the teacher-forced FlowMap baseline TF-sCM (84.04 and 83.78 vs. 82.94). Crucially, avoiding DMD preserves sample diversity (Figure 9): our π-Flow students surpass every corresponding SF-DMD student under V-JEPA 2 and four out of five under VideoMAE V2, and with the Causal Forcing teacher even surpass the 64-NFE teacher itself (0.0964 vs. 0.0882 in V-JEPA 2 cosine distance).

Long-video generation Table 2 evaluates the chunk-wise students on 30 s rollouts, and Figure 10 shows qualitative comparisons. With the Causal-rCM teacher, ours (π-Flow) achieves both higher overall quality (62.13 AQ and 67.41 IQ vs. 61.41 and 65.92 for Causal-rCM) and smaller quality degradation over 30 s (ΔIQ of -1.64 vs. -3.28, and ΔAQ of -3.83 vs. -4.06). With the Causal Forcing teacher, ours (π-Flow) matches the high dynamic degree of Causal Forcing (82.1 vs. 81.7) while exhibiting a larger drop in imaging quality (-4.94 vs. -1.44); Section 6.3 isolates how much anti-drifting guidance reduces this drift.

Table 2. **30 s generation.** VBench-Long on 30 s videos of the chunk-wise 4-NFE students of Table 1, generated with Deep Forcing and averaged over five seeds (Section 6.1): dynamic degree (DD), aesthetic quality (AQ), imaging quality (IQ), and ΔIQ and ΔAQ, the changes in imaging and aesthetic quality from the start to the end of the video. Colors follow Table 1; all dimensions are in the supplementary material.

Method	DD ↑	AQ ↑	IQ ↑	ΔIQ ↑	ΔAQ ↑
Self Forcing†	60.8	61.18	69.30	−0.76	−2.00
Causal Forcing teacher					
Causal Forcing†	81.7	58.71	68.92	−1.44	−5.59
Ours (π-Flow)	82.1	57.92	64.62	−4.94	−6.80
Ours (dCM)	49.4	54.89	62.04	−9.01	−11.56
Causal-rCM teacher					
Causal-rCM†	65.6	61.41	65.92	−3.28	−4.06
Causal-rCM, TF-sCM†	48.6	58.80	65.29	−3.10	−5.08
Ours (π-Flow)	41.1	62.13	67.41	−1.64	−3.83
Ours (dCM)	41.3	60.27	68.25	−1.97	−4.01

Table 3. **Contribution of each guidance method.** π-Flow students evaluated after Stage 2. Ours is the π-Flow student of Table 1, and removing both guidance methods leaves CFG alone in both stages. Results are colored from dark to light by rank: best, second, third and the rest.

Variant	VBench ↑
Ours	84.76
− anti-drifting guidance	84.35
− motion-aware guidance	84.19
− both (CFG only)	81.61

Table 4. **Stage 1 guidance.** Students evaluated after Stage 1. Fixed-strength rows use skip-layer guidance with the same strength u at every patch; $u = 0.5$ is the base strength $\bar{u}$ of motion-aware guidance and $u = 0.61$ its mean strength during training. Removing skip-layer guidance leaves CFG. Colors rank the π-Flow rows as in Table 3.

Method	Teacher guidance	VBench ↑
π-Flow	Motion-aware (ours)	84.01
	fixed $u = 0.5$	83.54
	fixed $u = 0.61$	83.69
	fixed $u = 1$	83.43
	− skip-layer guidance	83.66
dCM	Motion-aware (ours)	82.88
	− skip-layer guidance	82.64

6.3. Ablation Studies

We ablate with the chunk-wise Causal Forcing teacher; the same ablations with the Causal-rCM teacher are in the supplementary material. All students generate with 4 NFE, and VBench is the Total score averaged over five seeds.

Table 5. **Stage 2 guidance.** Students evaluated after Stage 2. ΔIQ and ΔAQ are the changes in imaging and aesthetic quality over 30 s videos (Section 6.1; higher is better), and VBench is measured on 5 s videos. All rows retain motion-aware guidance in both stages, and removing anti-drifting guidance sets $\gamma = 1$. Colors rank the π-Flow rows as in Table 3.

		30 s		5 s
Method	Teacher guidance	ΔIQ ↑	ΔAQ ↑	VBench ↑
π-Flow	Anti-drifting (ours)	−4.94	−6.80	84.76
	− anti-drifting guidance	−7.79	−9.49	84.35
dCM	Anti-drifting (ours)	−9.01	−11.56	83.70
	− anti-drifting guidance	−10.00	−16.51	83.59

Table 6. **Update frequency of the drift model in anti-drifting guidance.** π-Flow students evaluated after Stage 2 across drift-model updates per student update n at guidance scale $\gamma = 5$; $\gamma = 1$ removes anti-drifting guidance. Diversity is the cosine distance of Table 1 (higher is more diverse). Colors rank the rows as in Table 3.

n	γ	VBench ↑	V-JEPA 2 ↑	VideoMAE V2 ↑
–	1	84.35	0.0905	0.0239
1	5	84.56	0.0902	0.0252
5	5 (ours)	84.76	0.0964	0.0281

Components Table 3 ablates our two guidance methods on π-Flow. Starting from the CFG-only baseline without either guidance yields 81.61 VBench. Adding anti-drifting guidance in Stage 2 (84.19) or motion-aware guidance in both stages (84.35) each improves quality by more than 2.5 points, and combining both achieves the best score (84.76), as motion-aware guidance smooths the teacher's velocity field in moving regions within each chunk while anti-drifting guidance counteracts error accumulation across chunks.

Stage 1 Table 4 evaluates the students directly after teacher forcing. With π-Flow, uniform skip-layer guidance at fixed strengths $u = 0.5$ (83.54) and $u = 1$ (83.43) lowers VBench below CFG alone (83.66) and barely improves at the matched mean strength $u = 0.61$ (83.69), whereas motion-aware guidance reaches 84.01 (and raises dCM from 82.64 to 82.88). Unlike uniform skip-layer guidance, which perturbs already-certain static regions and saturates early (Figure 4(b)), motion-aware guidance concentrates skip-layer guidance on high-uncertainty moving patches.

Stage 2 Table 5 evaluates anti-drifting guidance after self-forcing. On 30 s rollouts (Figure 11), anti-drifting guidance reduces quality degradation for π-Flow from −7.79 to



(a) "… a young white rabbit … reading a newspaper"

(b) "… a corgi in tropical Maui …"

Figure 11. **Anti-drifting guidance on 30 s videos.** Frames at 0, 10, 20 and 30 s from ours (π-Flow) with the Causal Forcing teacher, trained without and with anti-drifting guidance (Table 5), generated with Deep Forcing from the same prompt and seed (prompts abbreviated).

-4.94 in ΔIQ and from -9.49 to -6.80 in ΔAQ while raising 5 s VBench from 84.35 to 84.76 (and with the Causal-rCM teacher, from -3.88 to -1.64 in ΔIQ and from -6.41 to -3.83 in ΔAQ; supplementary material); for dCM, it improves ΔAQ from -16.51 to -11.56 and ΔIQ from -10.00 to -9.01 with similar 5 s VBench (83.59 to 83.70). Table 6 varies the drift-model update frequency n at $\gamma = 5$: a single update ($n = 1$) already raises VBench from 84.35 to 84.56, and $n = 5$ updates allow the drift model to track the evolving student more closely, reaching 84.76 VBench while increasing V-JEPA 2 and VideoMAE V2 cosine distances from 0.0905 and 0.0239 to 0.0964 and 0.0281 (see the supplementary material for sweeps over γ on 5 s and 30 s rollouts).

7. Discussion and Conclusion

Limitations Because FlowMap distillation follows the AR teacher's trajectories, the teacher bounds the student's capabilities: while achieving the highest VBench Quality scores, our 1.3B-distilled students have lower Semantic scores than SF-DMD students distilled from 14B teachers, and with the Causal-rCM teacher a lower dynamic degree (Table 1). In 30 s generation, anti-drifting guidance substantially reduces rollout drift (Table 5), though with the Causal Forcing teacher the imaging quality of ours (π-Flow) still drops more than that of Causal Forcing (-4.94 vs. -1.44; Table 2). Finally, maintaining the drift model requires n additional updates per student update during Stage 2.

Conclusion We have shown that distribution matching distillation (DMD) is not required to distill autoregressive video models into few-step students. Two properties of causal AR generation have previously limited two-stage FlowMap distillation: in Stage 1 (teacher forcing), the history after the first chunk fixes the scene's appearance so that uncertainty concentrates in moving regions where the teacher's velocity field changes sharply and text-conditioned CFG fails to smooth it; and in Stage 2 (self-forcing), matching the AR teacher's conditional distribution given a student rollout continues the history's drift rather than pulling the sequence back toward the data distribution. Motion-aware guidance and anti-drifting guidance address these two obstacles by modifying only the teacher's velocity field, making them directly applicable to both π-Flow and consistency distillation. Distilled from 1.3B AR teachers without any DMD stage, the resulting 4-step FlowMap students match SF-DMD students distilled from 14B teachers on VBench while generating substantially more diverse videos and mitigating drift in 30 s rollouts.

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